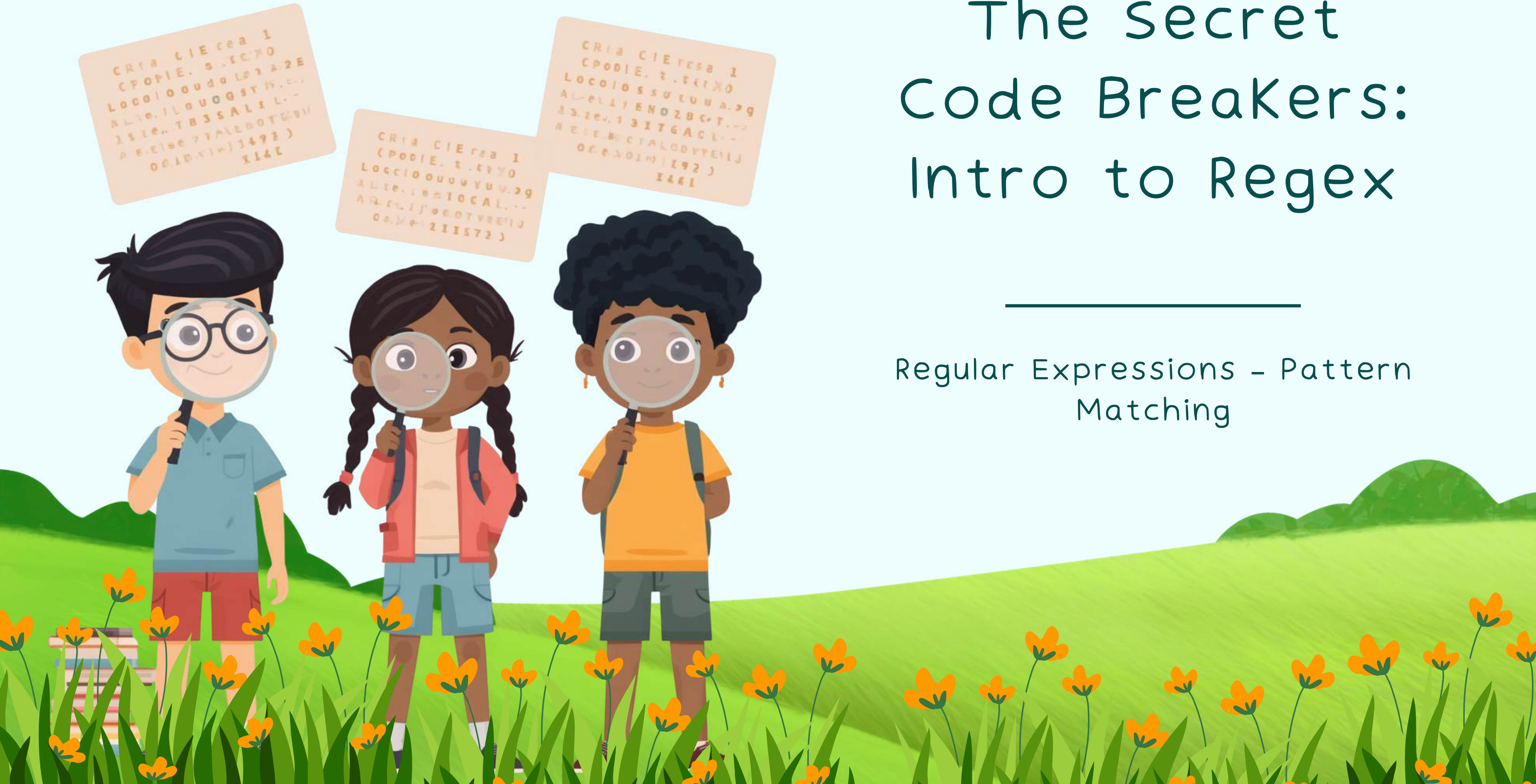


The Secret Code Breakers: Intro to Regex

Regular Expressions – Pattern
Matching



The Challenge

Secret Codes are special search patterns that find things even with missing pieces

The Solution

Today we become Data Detectives solving the mystery!

The Library Mystery

Can You Crack the Code?

Imagine you find a book on the shelf, but its title is partly torn... You can only see some letters. Can you guess the missing ones? This is exactly what Secret Codes (Regex) help us do!



Regex is like a secret code to help computers find exactly what we want. It's a superpower for data detectives!

What is Regex?

A secret code for computers!

02

Normal search: looks for exact words only

Regex search: looks for patterns (like missing letters!)

Regex helps us find things even with incomplete info!



Dot (.) means "any single character"

Example: "R.m" matches Ram, Rim, Rom

It's like a wildcard that fits any letter!

05

Plus (+) means "one or more of the previous"

Example: "go+l" matches gol, gool, goool

The letter keeps repeating one or more times!

Meet the Secret Symbols

– Dot (.) and Plus (+)

Basic metacharacters for pattern matching



Guided Practice

The Postman's Helper

Your Task:

Find all valid 6-digit PIN codes in this text!

PIN codes start with 1-9 and have exactly 6 digits.

The Regex Pattern:

`[1-9][0-9]{5}`

Sample Text:

"Send to 400001 Mumbai, 110001
Delhi, 12345 invalid, 560034
Bangalore"

Matches Found:

✓ 400001 ✓ 110001 ✓ 560034
x 12345 (only 5 digits!)



Pause & Think #1

Time to reflect and discuss!

Why do you think computers need patterns like Regex?
Think about how many emails, phone numbers, and codes exist in the world!

Share your thoughts with your classmates. There are no wrong answers – just curious minds exploring!



Asterisk (*)* means "zero or more of the previous character"

Example: colour matches "color" and "colour"

Great for finding words with optional letters!

More Secret Symbols – Asterisk (*) and Character Classes []

Unlocking more powerful pattern matching tools!

[]

Character Classes [0-9] means any digit from 0 to 9

Mobile numbers: +91 followed by 10 digits

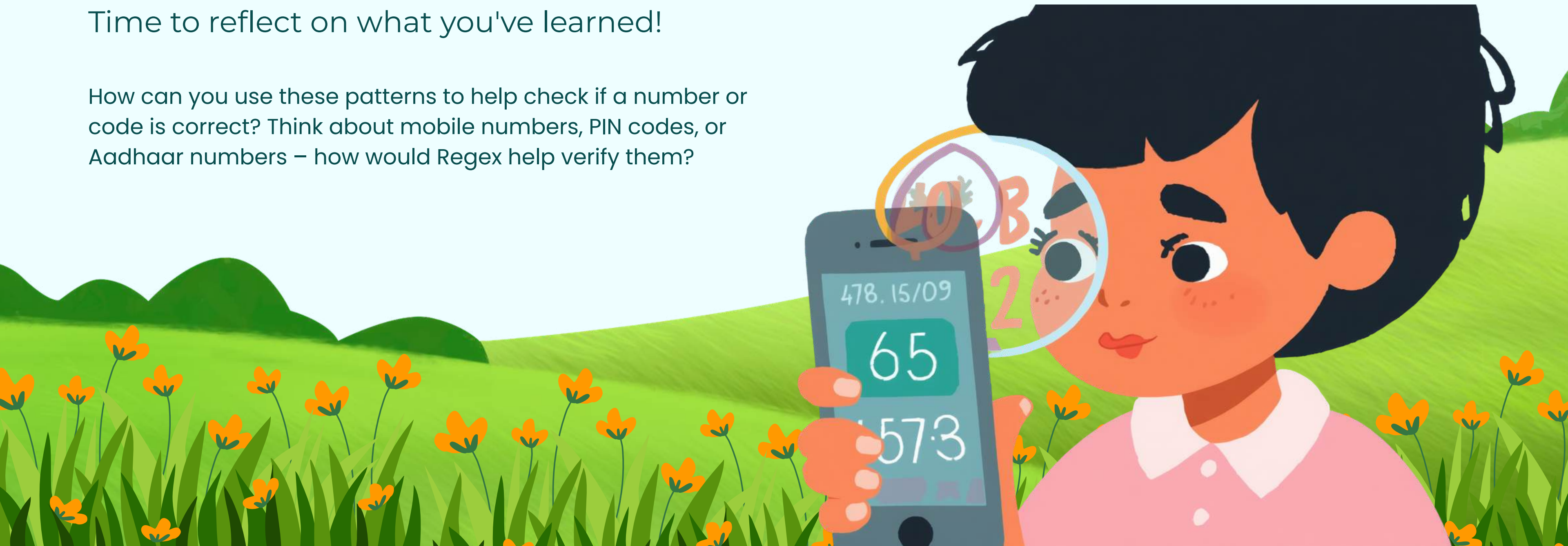
Aadhaar: 12-digit unique ID number



Pause & Think #2

Time to reflect on what you've learned!

How can you use these patterns to help check if a number or code is correct? Think about mobile numbers, PIN codes, or Aadhaar numbers – how would Regex help verify them?



Your Task

Hint: Use dot (.) to replace the letter that changes

Secret Tip

Pattern: Ind.a matches both spellings!

Independent Challenge

Find "India" Even if Misspelled!

Create a pattern to find "India" or "Indya" in any text. The dot (.) is your secret weapon – it matches any single character. So Ind.a will match both India and Indya!



Dot (.) – Matches any single character

Example: "R.m" finds Ram, Rim, Rom

Plus (+) – One or more of previous

Example: "a+" finds a, aa, aaa

Regex Cheat Sheet

Your Detective's Quick
Reference Guide

Asterisk (*) – Zero or more of previous

Example: "go" finds g, go, goo

Character Class [0-9] – Any digit

Example: "[1-9][0-9]{5}" finds PIN codes



Quick Quiz – What Will This Pattern Match?

Test Your Secret Code Skills!

What does R.m match?

- a) Only "Ram"
- b) "Ram", "Rim", "Rom"
- c) "Room"

Answer: b

What does `[1-9][0-9]{5}` find?

- a) Any 6 letters
- b) Valid PIN codes
- c) Phone numbers

Answer: b

What does `a+` match?

- a) Zero "a" letters
- b) One or more "a" letters
- c) Exactly one "a"

Answer: b

What does `[0-9]*` match?

- a) Zero or more digits
- b) Only one digit
- c) Letters only

Answer: a





Thanks for being
great Secret
Code Breakers!

Keep practicing and stay
curious about Regex patterns!